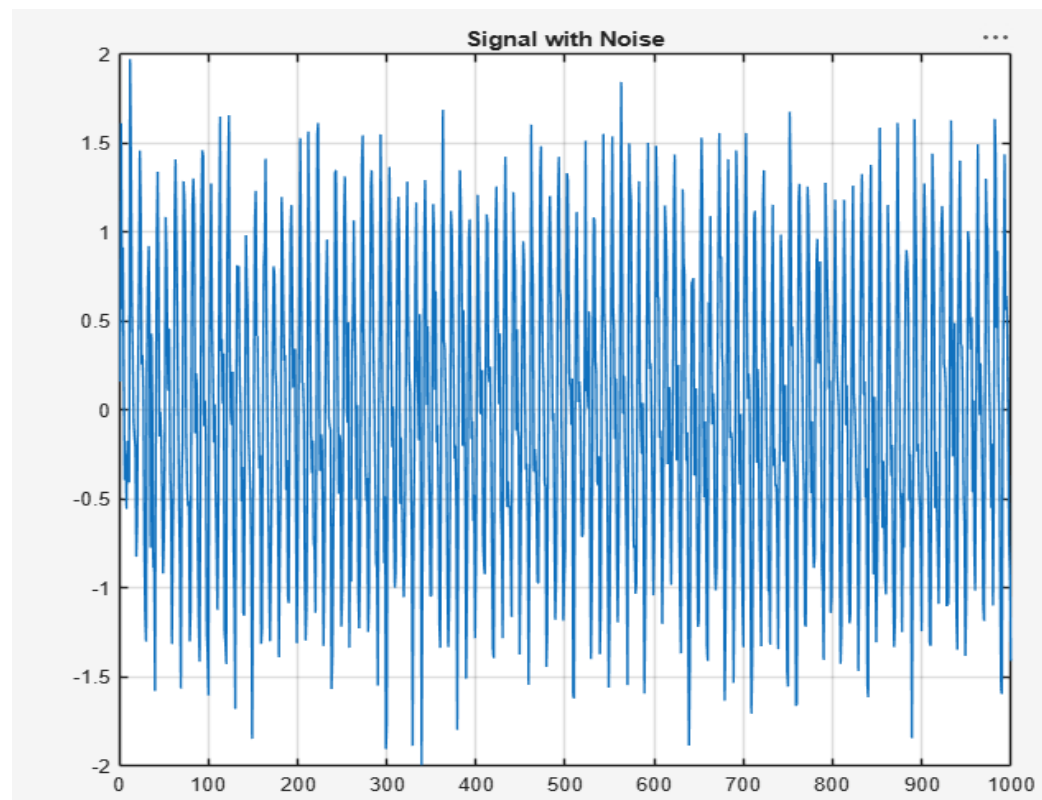


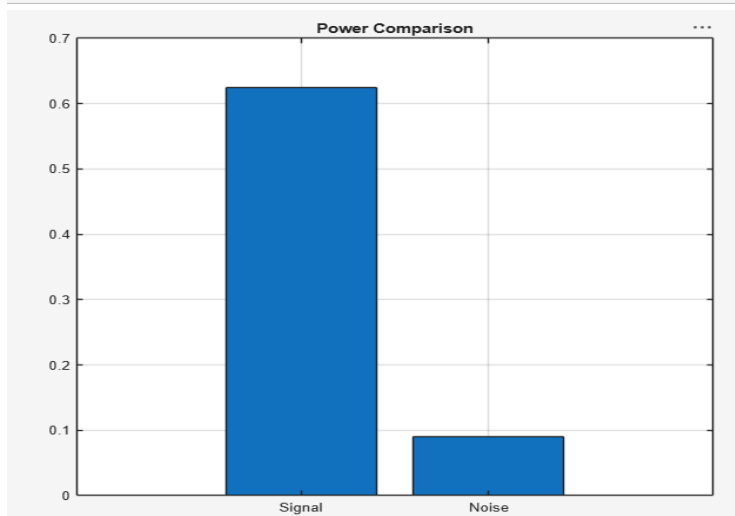
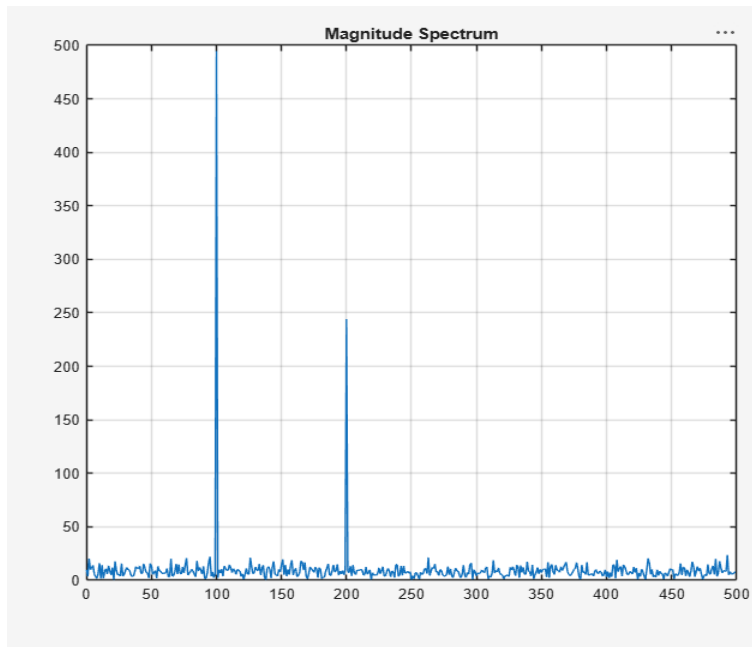
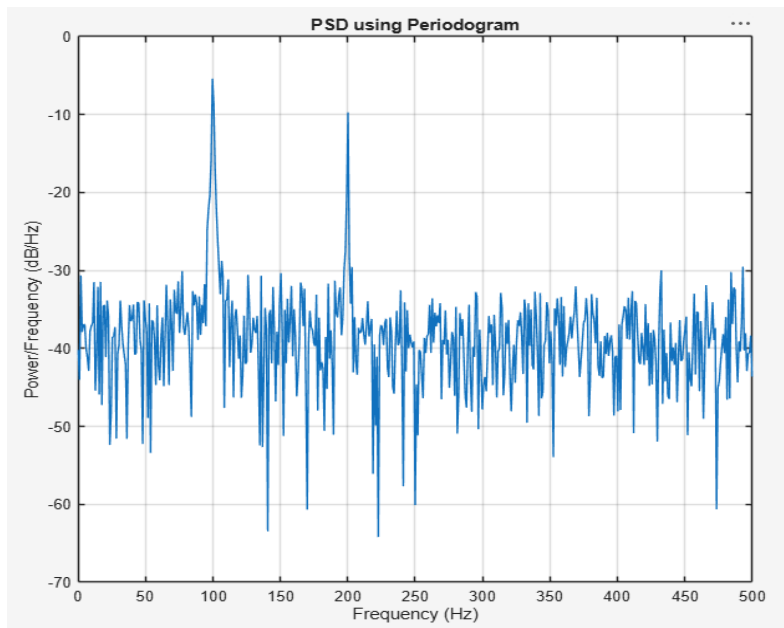
7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals

Periodogram Method in MATLAB

```
1  clc; clear; close all
2  fs=1000; t=(0:999)/fs;
3  s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
4  n=0.3*randn(size(t));
5  x=s+n;
6  figure, plot(x), grid on
7  title('Signal with Noise')
8  figure, periodogram(x,[],[],fs)
9  title('PSD using Periodogram')
10 figure
11 X=abs(fft(x));
12 f=(0:length(X)-1)*fs/length(X);
13 plot(f(1:500),X(1:500)), grid on
14 title('Magnitude Spectrum')
15 figure
16 bar([mean(s.^2) var(n)])
17 set(gca,'XTickLabel',{'Signal','Noise'})
18 title('Power Comparison')
19 grid on
```

Output:





7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, Hanning Windows in MATLAB

1	clc; clear; close all;	
2	% Signal	
3	fs = 1000; N = 256; t = (0:N-1)/fs;	
4	x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);	
5	NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);	
6	% Windows	
7	w = {rectwin(N), hamming(N), hann(N)};	
8	names = {'Rectangular', 'Hamming', 'Hanning'};	
9	% Processing	
10	for i = 1:3	
11	xw = x(:).*w{i};	
12	X = fft(xw,NFFT);	
13	P{i} = abs(X(1:NFFT/2));	
14	~P{i} = P{i}/max(P{i});	
15	end	
16	%% Individual Spectrums	
17	figure;	
18	for i = 1:3	
19	subplot(3,1,i);	
20	plot(f,20*log10(P{i}), 'LineWidth',1.5);	
21	title(['Spectrum using ' names{i} ' Window']);	
22	xlim([0 250]); grid on;	
23	end	
24	%% Comparison	
25	figure;	
26	plot(f,20*log10(P{1}), 'k', f, 20*log10(P{2}), 'r', f, 20*log10(P{3}), 'b', 'LineWidth',1.5);	
27	legend(names); title('Window Comparison'); xlim([50 170]); grid on;	
28	%% Window Shapes	
29	figure;	
30	for i = 1:3	
31	subplot(3,1,i); plot(w{i}, 'LineWidth',1.5);	
32	title([names{i} ' Window']); grid on;	
33	end	
34	%% Peak Detection	
35	for i = 1:3	
36	[pks{i}, loc{i}] = findpeaks(P{i}, f, 'MinPeakHeight', 0.3);	
37	fprintf('\n%s Window Frequencies:\n', names{i});	
38	disp(loc{i});	
39	end	
40	%% Peak Marking (Hamming)	
41	figure;	
42	plot(f, 20*log10(P{2}), 'LineWidth', 1.5); hold on;	
43	plot(loc{2}, 20*log10(pks{2}), 'ro', 'MarkerFaceColor', 'r');	
44	title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;	
45	%% Main Peaks	
46	fprintf('\nMain Spectral Peaks:\n');	
47	for i = 1:3	
48	[~, idx] = max(P{i});	
49	fprintf('%s Window Peak = %.2f Hz\n', names{i}, f(idx));	
50	end	

Output:

